

Foundation Models & Embeddings

Lara J. Martin (she/they)

<https://laramartin.net/neurosymbolic-text-gen/>

Plagiarism example slides modified from Dr. Frank Ferraro

Learning Objectives

Try common prompting techniques like chain-of-thought

Recognize useful encoder-only, encoder-decoder, and decoder-only models

Understand the use & creation of dense vector embeddings

Calculate the distance between vector embeddings

Differentiate between encoder model embeddings and older dense embeddings

In-Class Activity

Think of something you're an expert in. It can be anything!

Pick an LLM that's hosted online such as ChatGPT (<https://chatgpt.com/>) or Claude (<https://claude.ai>).

Ask your LLM to give you information about that topic. Ask in different ways about different things and use different prompting techniques.

What does it do well with?

What does it not do well with?

Some Prompting Techniques:

Few-shot

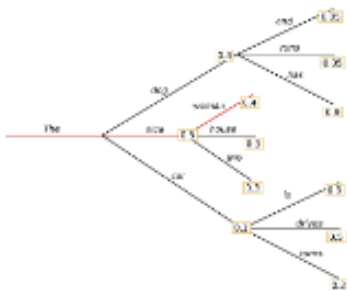
Zero-shot

Chain of thought

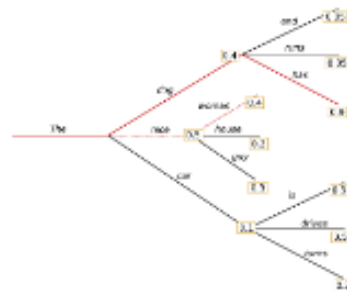
Decomposition

Self-Criticism

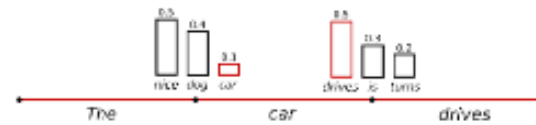
Review: Sampling



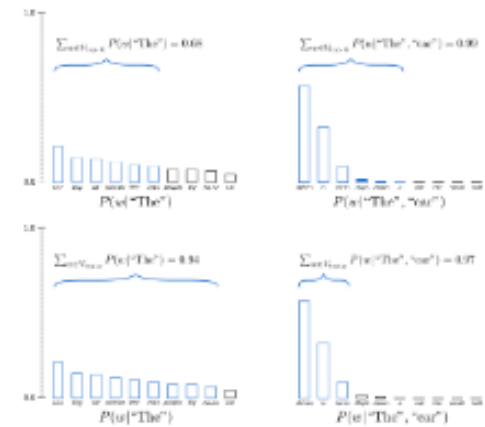
Greedy



Beam Search



Random Sampling



Top-K/P

Review

What's the difference between finetuning and prompting?

What's the difference between zero-shot and few-shot prompting?

Review: Tricks of the Trade

Instruction-tuned models like GPT-3.5 and Mistral-7B-Instruct like to be given a “role” first (e.g., “You are a helpful writing assistant.”)

The more defined the task, the better

- More details
- One thing to do at a time

LLMs are overly confident (like people on the internet)

- To “objectively” have the model evaluate something, you should have another instance judge

Chain-of-thought prompting helps models come up with better answers

They will “Yes and...” your prompt

LLM Vocabulary

Finetuning: Training an LLM more to adapt it to your task

Pre-training: The training before finetuning; creating a foundation model

Prompting: Getting the output you want by just changing the input; the model doesn't change

Zero-shot prompting: Prompting without examples

Few-shot prompting: Prompting with a few examples

Prompt engineering: Figuring out the right prompt for a task

Prompt optimization: Automated prompt engineering

Prompt tuning: Automated creation of *latent* prompt (e.g., vector representation)

Foundation Models

Types of Foundation Models

Encoder Only

Decoder Only

Encoder-Decoder Models



Denotes what they use
during pre-training

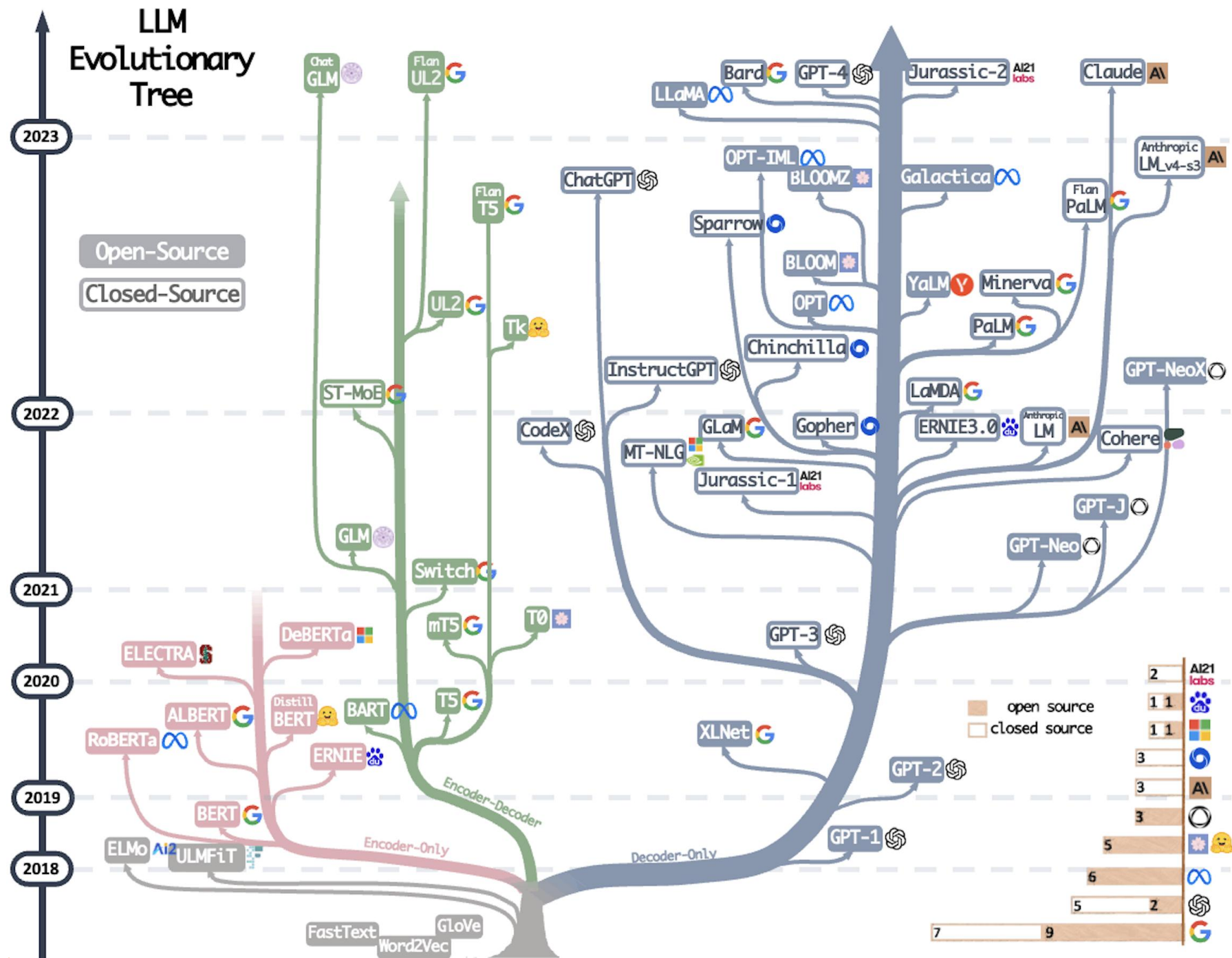
What is a foundation model?

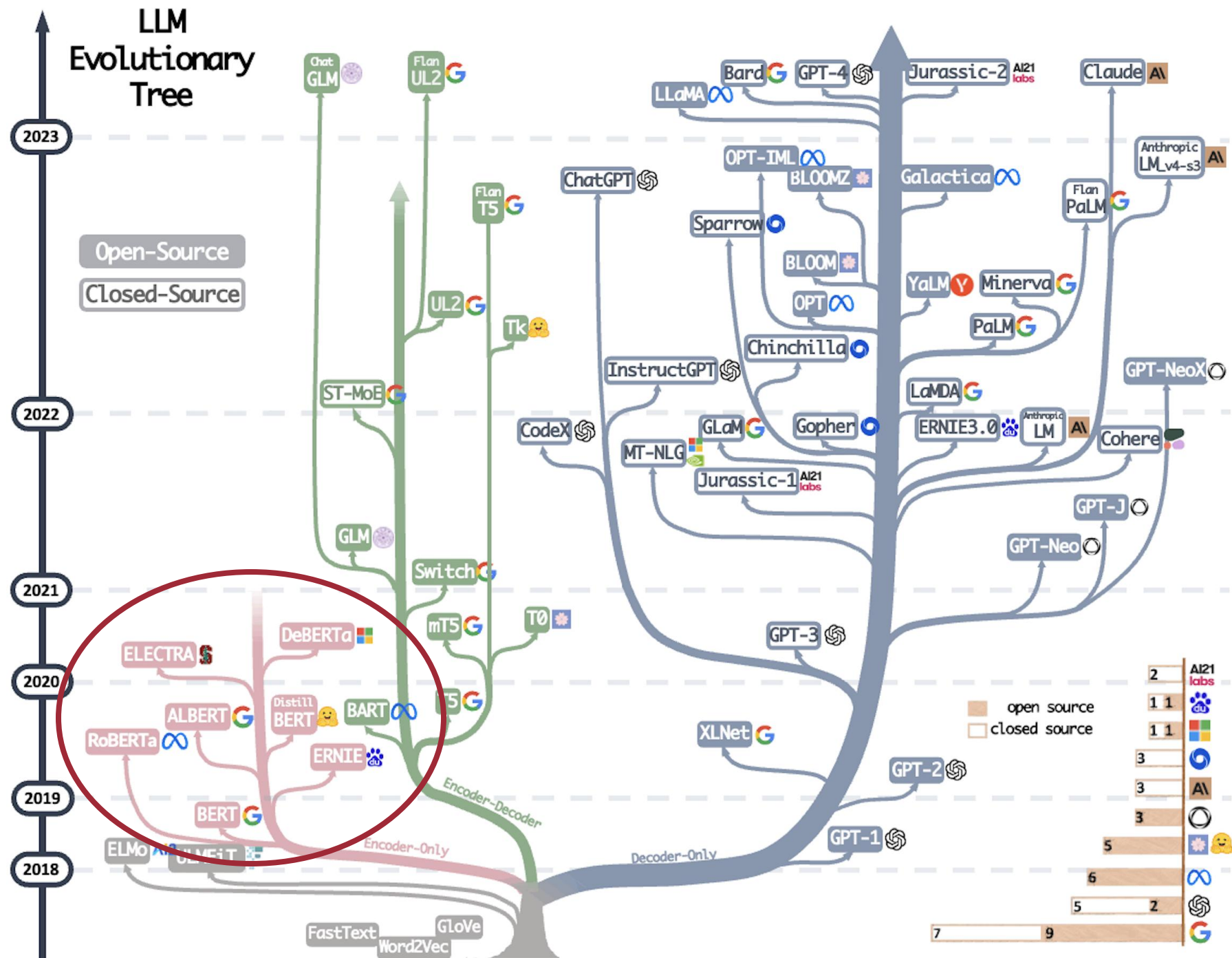
A model that captures “foundational” or core information about a modality (e.g., text, speech, images)

Pretrained on a large amount of data & able to be finetuned on a particular task

Self-supervised

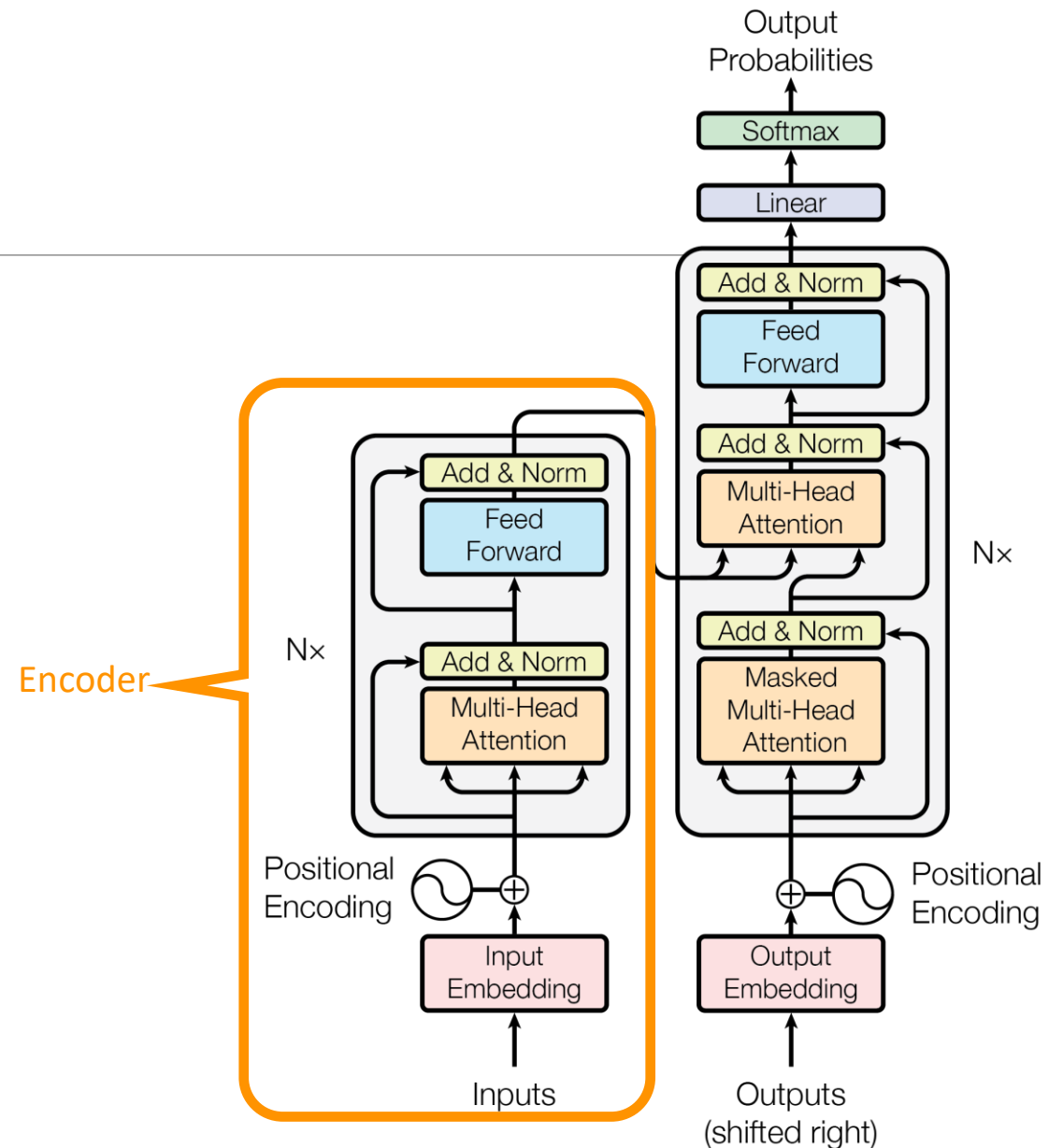
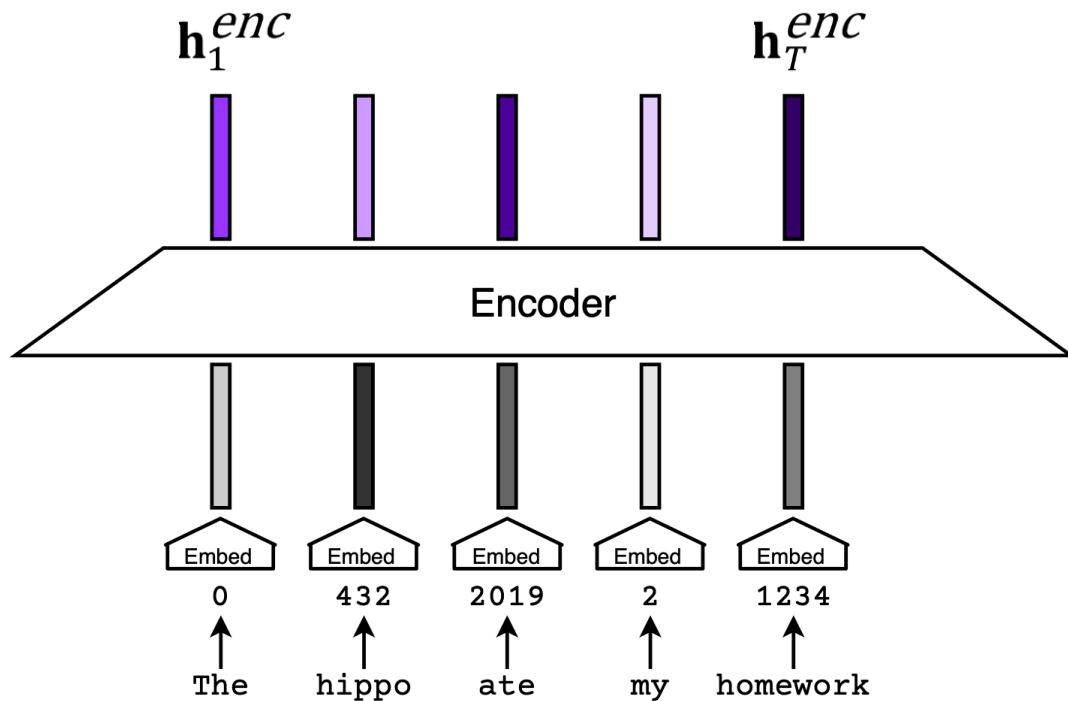
All non-finetuned pretrained large language models (LLMs) are foundation models



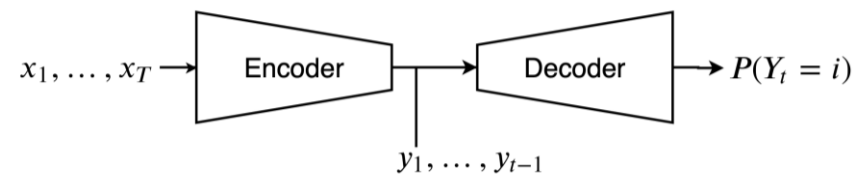




Encoder-only models

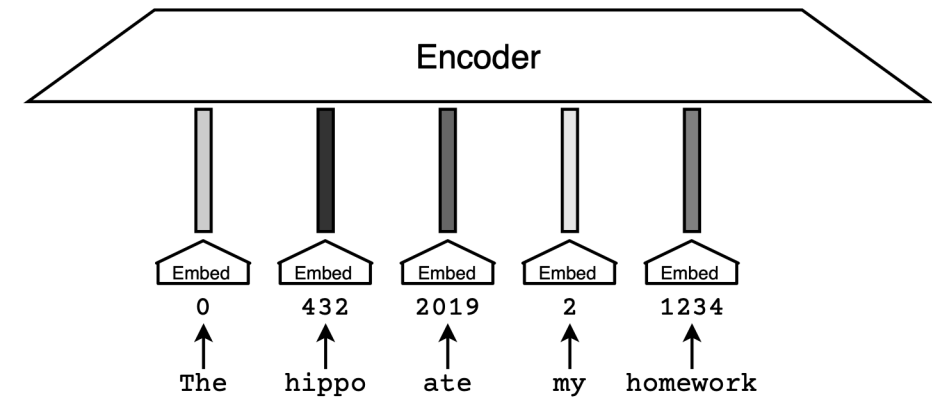
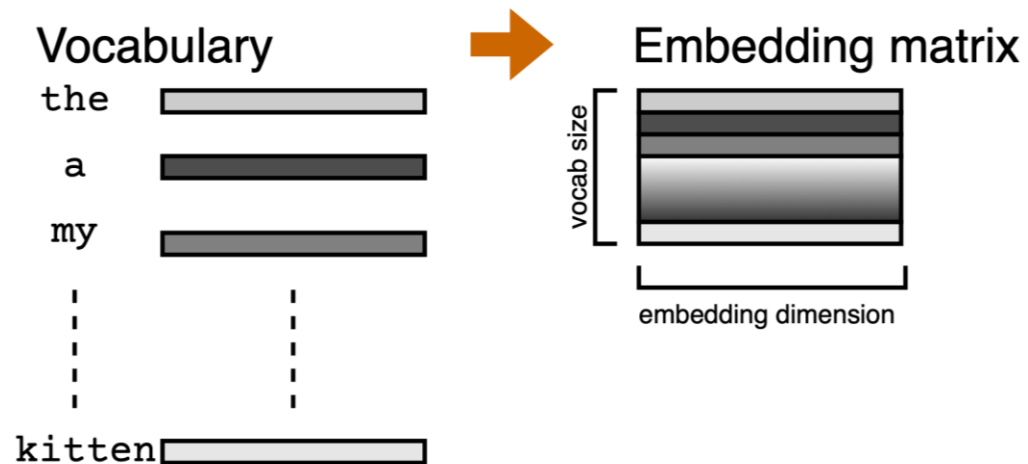


Word Embeddings



Review: Inputs to the Encoder

The encoder takes as input the embeddings corresponding to each token in the sequence.



How have we represented words?

Each word is a distinct item

- Bijection between the strings and unique integer ids:
- "cat" --> 3, "kitten" --> 792 "dog" --> 17394
- Are "cat" and "kitten" similar?

Equivalently: "One-hot" encoding

- Represent each word type w with a vector the size of the vocabulary
- This vector has $V-1$ zero entries, and 1 non-zero (one) entry

One-Hot Encoding Example

Let our vocab be {a, cat, saw, mouse, happy}

$V = \# \text{ types} = 5$

Assign:

a	4
cat	2
saw	3
mouse	0
happy	1

How do we
represent “cat?”

$$e_{\text{cat}} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

How do we
represent
“happy?”

$$e_{\text{happy}} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

The Fragility of One-Hot Encodings

Case Study: Plagiarism Detector

Given two documents x_1, x_2 , predict $y = 1$ (plagiarized) or $y = 0$ (not plagiarized)

What is/are the:

Method/steps for predicting?

General formulation?

Features?



There's no way you'll
catch me!

Plagiarism Detection: Word Similarity?

MAINFRAMES

Mainframes **are primarily** referred to large computers with **rapid**, advanced processing capabilities that **can execute and** perform tasks **equivalent to many** Personal Computers (PCs) machines **networked together**. It is **characterized with high quantity** Random Access Memory (RAM), very large secondary storage devices, and **high-speed** processors to cater for the needs of the computers under its service.

Consisting of advanced components, mainframes have the capability of running multiple large applications required by **many and** most enterprises **and organizations**. **This is** one of its advantages. Mainframes are also suitable to cater for those applications **(programs)** or files that are of very **high** demand by its users (clients). Examples of **such organizations and enterprises using mainframes are** online shopping websites **such as**

MAINFRAMES

Mainframes **usually are** referred those computers with **fast**, advanced processing capabilities that **could perform by itself** tasks **that may require a lot of** Personal Computers (PC) Machines. **Usually mainframes would have lots of** RAMs, very large secondary storage devices, and **very fast** processors to cater for the needs of those computers under its service.

Due to the advanced components mainframes have, **these computers** have the capability of running multiple large applications required by most enterprises, **which is** one of its advantage. Mainframes are also suitable to cater for those applications or files that are of very **large** demand by its users (clients). Examples of these **include** the large online shopping websites **-i.e. : Ebay, Amazon, Microsoft, etc.**

Case Study: Plagiarism Detector (Feature Example)

Given two documents x_1, x_2 , predict $y = 1$ (plagiarized) or $y = 0$ (not plagiarized)

Intuition: documents are more likely to be plagiarized if they have words in common

$$f_{\text{any-common-word, Plag.}}(x_1, x_2) = ???$$

$$f_{\langle \text{word } v \rangle, \text{Plag.}}(x_1, x_2) = ???$$



Yes, but surely some words will be in common... these features won't catch phrases!

Example:

$$e_{\text{cat}} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

Case Study: Plagiarism Detector (Feature Example)

Given two documents x_1, x_2 , predict $y = 1$ (plagiarized) or $y = 0$ (not plagiarized)

Intuition: documents are more likely to be plagiarized if they have words in common

n # adjacent words

$$f_{\text{any-common-word, Plag.}}(x_1, x_2) = ???$$

$$f_{\langle \text{word } v \rangle, \text{Plag.}}(x_1, x_2) = ???$$

$$f_{\langle \text{ngram } Z \rangle, \text{Plag.}}(x_1, x_2) = ???$$

Example:

$$e_{\text{fluffy cat}} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$



No problem, I'll
just change
some words!

Case Study: Plagiarism Detector (Feature Example)

Given two documents x_1, x_2 , predict $y = 1$ (plagiarized) or $y = 0$ (not plagiarized)

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$$f_{\langle \text{ngram } Z \rangle, \text{Plag.}}(x_1, x_2) = ???$$

$$f_{\text{synonym-of-}\langle \text{word } v \rangle, \text{Plag.}}(x_1, x_2) = ???$$

Example:

$$e_{\text{feline}} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$



Okay... but there
are too many
possible synonym
n-grams!

Case Study: Plagiarism Detector (Feature Example)

Given two documents x_1, x_2 , predict $y = 1$ (plagiarized) or $y = 0$ (not plagiarized)

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$$f_{\text{synonym-of-}\langle \text{ngram } Z \rangle, \text{Plag.}}(x_1, x_2) = ???$$

What is the synonym of

$$e_{\text{fluffy cat}} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} ?$$



Hah, I win!

Word Embeddings

A dense, “low”-dimensional vector representation

Many values
are not 0 (or at
least less
sparse than
one-hot)

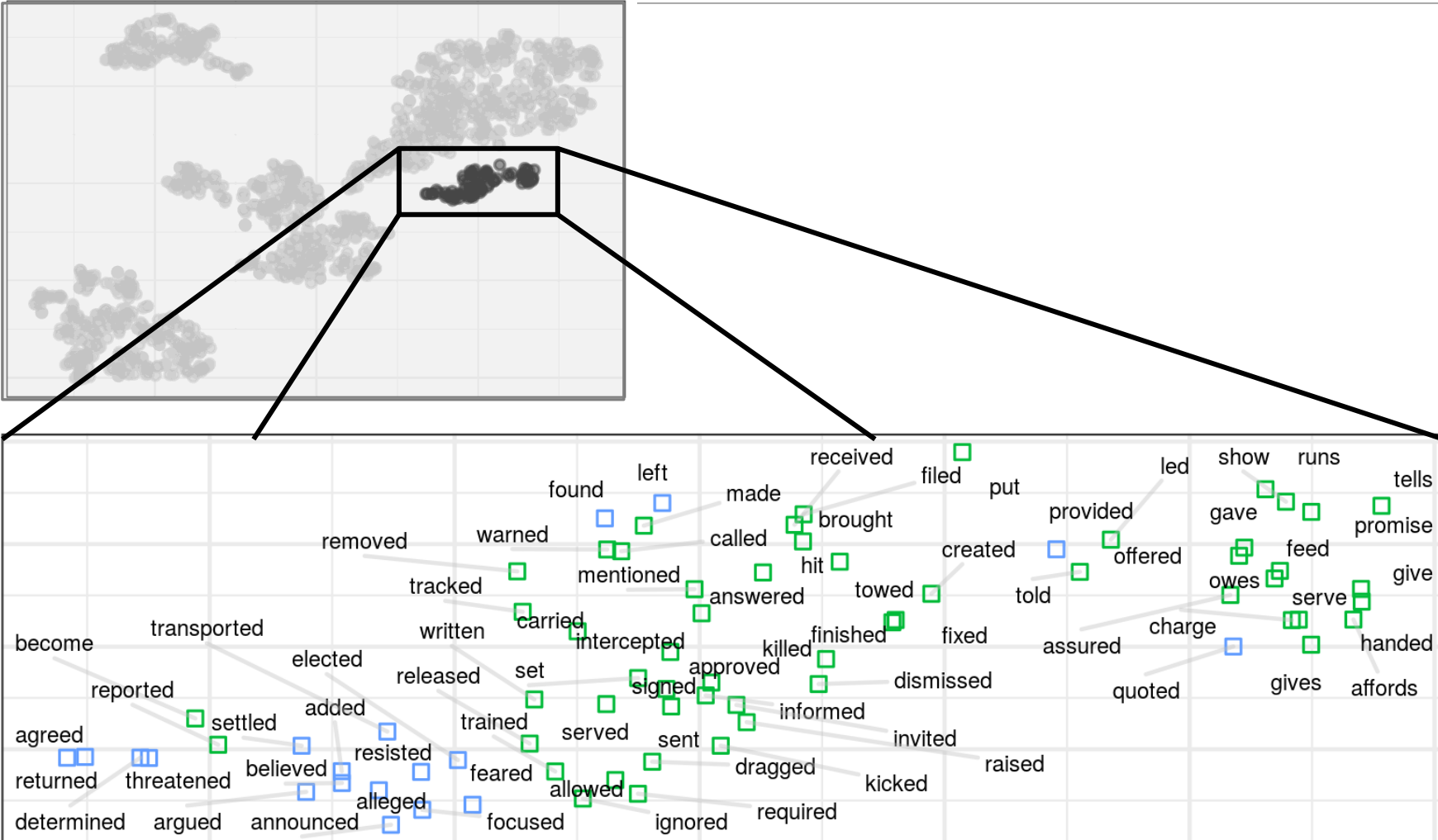
Up till ~2013: E could be
any size
2013-present: E << vocab

An E-dimensional
vector, often (but not
always) real-valued

These are also called

- **embeddings**
- **Continuous representations**
- **(word/sentence/...) vectors**
- **Vector-space models**

A Dense Representation

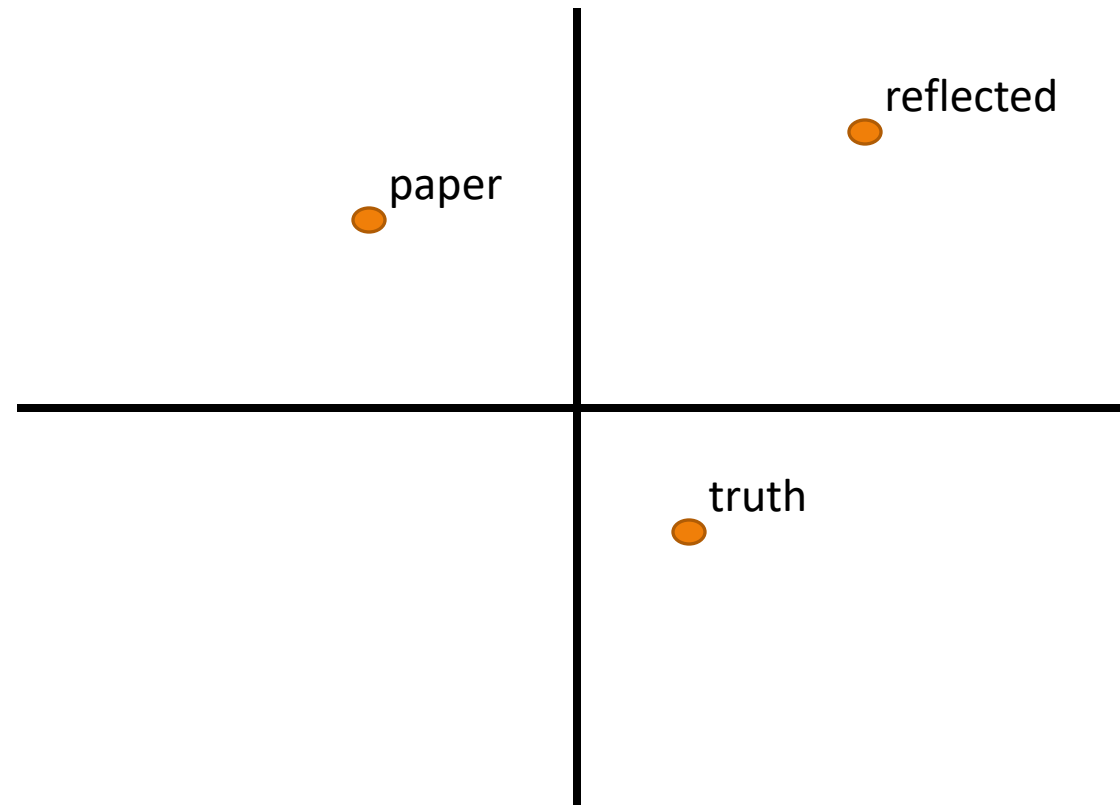


Continuous Meaning

The paper reflected the truth.

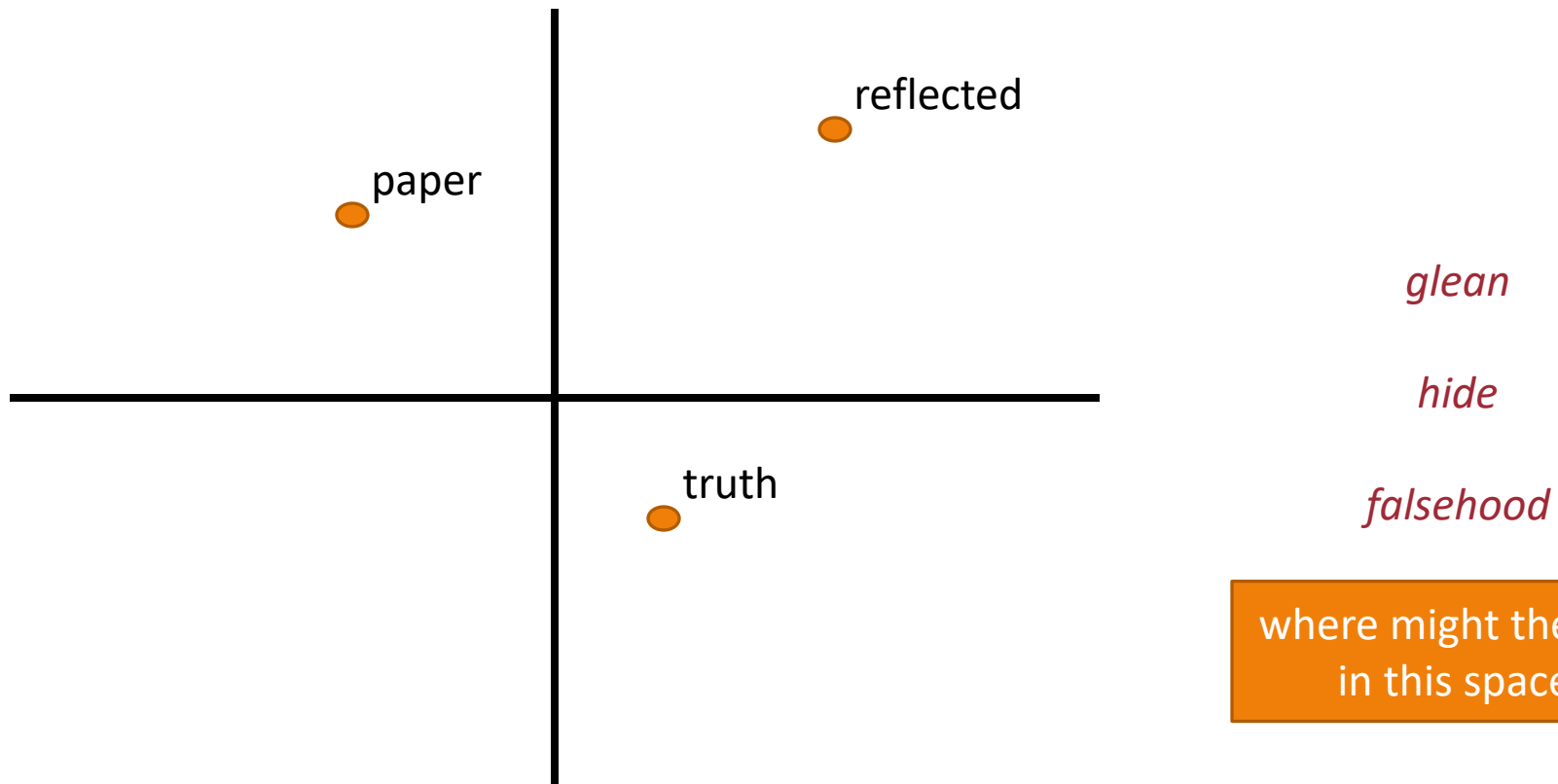
Continuous Meaning

The paper reflected the truth.



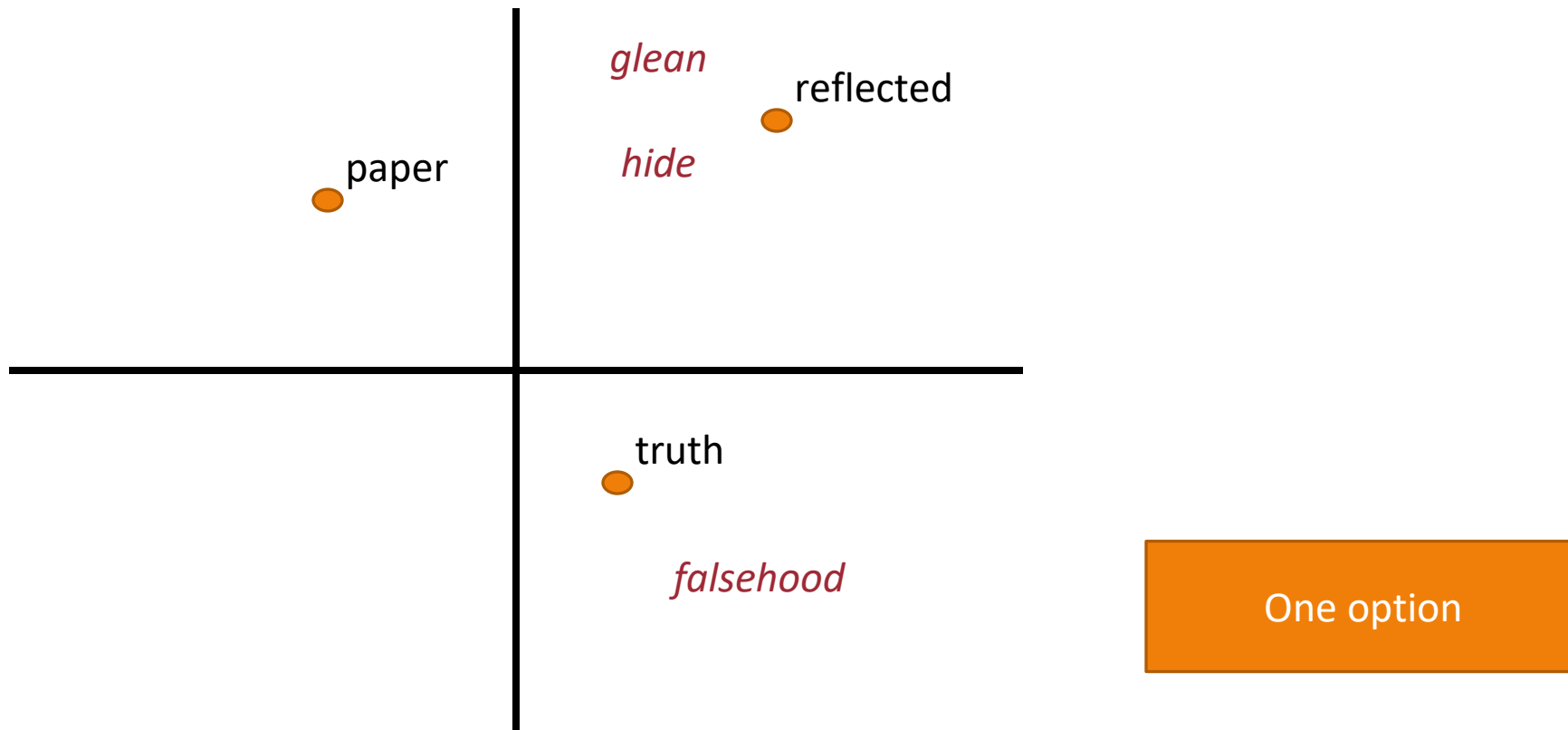
Continuous Meaning

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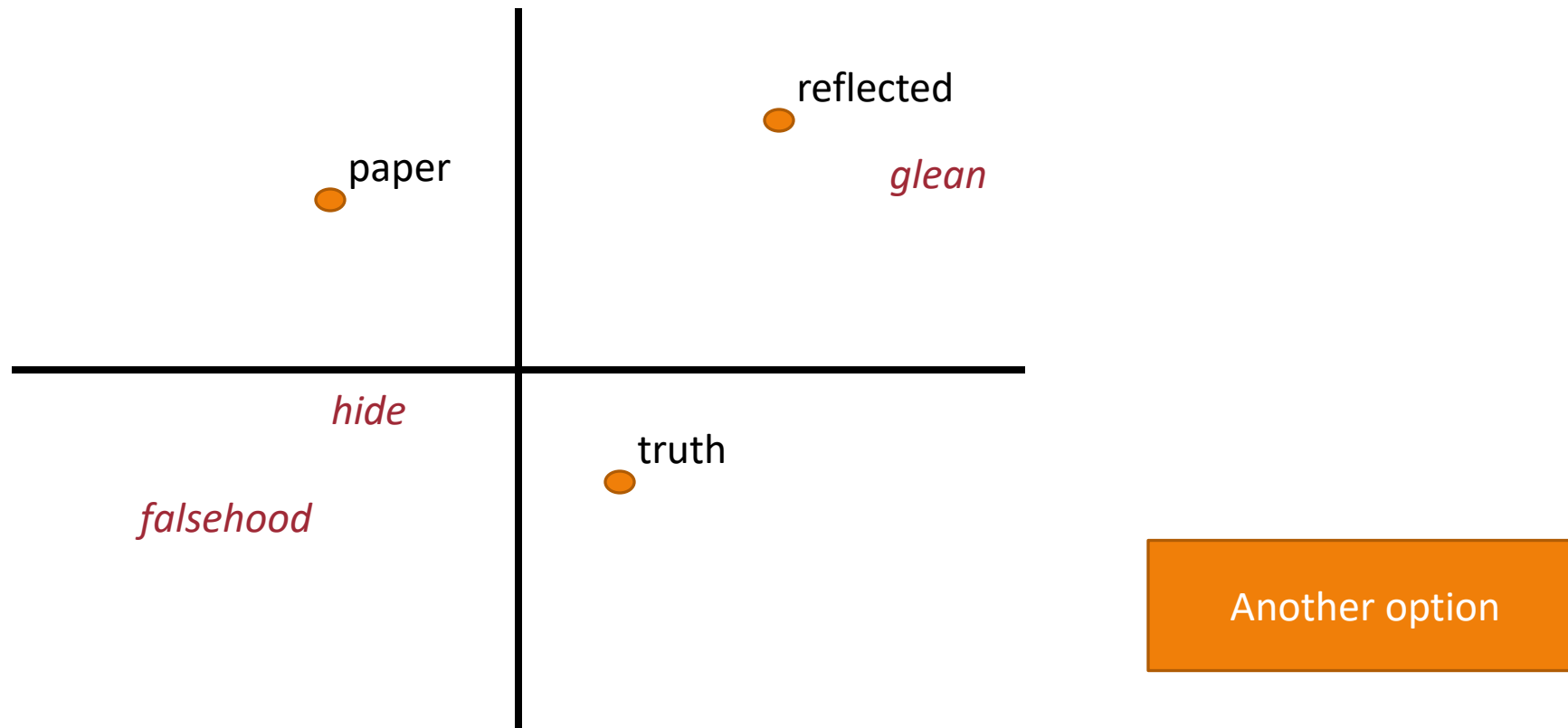
Continuous Meaning

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Continuous Meaning

The paper reflected the truth.



(Some) Properties of Embeddings

Capture “like” (similar) words



https://media3.giphy.com/media/3orif0M8U1E7NfpFzq/200_s.gif

target:	Redmond	Havel	ninjutsu	graffiti	capitulate
	Redmond Wash.	Vaclav Havel	ninja	spray paint	capitulation
	Redmond Washington	president Vaclav Havel	martial arts	grafitti	capitulated
	Microsoft	Velvet Revolution	swordsmanship	taggers	capitulating

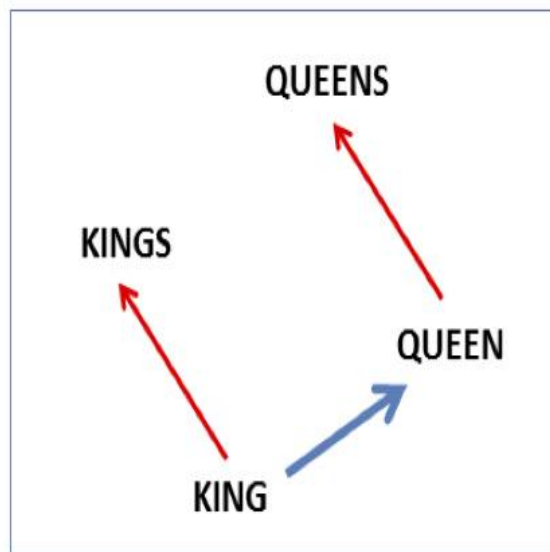
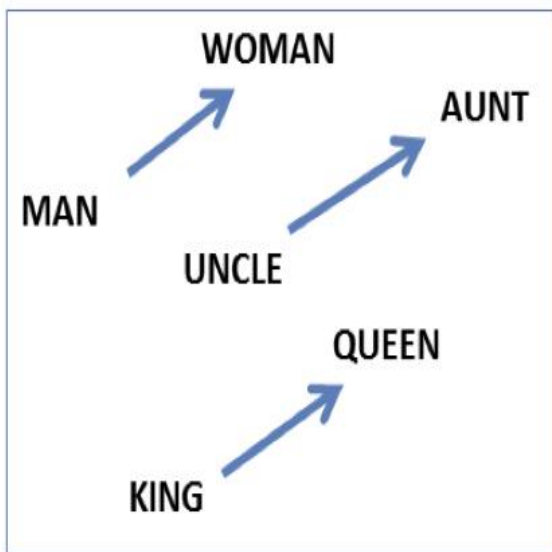
(Some) Properties of Embeddings



Capture “like” (similar) words

target:	Redmond	Havel	ninjutsu	graffiti	capitulate
	Redmond Wash.	Vaclav Havel	ninja	spray paint	capitulation
	Redmond Washington	president Vaclav Havel	martial arts	grafitti	capitulated
	Microsoft	Velvet Revolution	swordsmanship	taggers	capitulating

Capture relationships



$\text{vector}('king') - \text{vector}('man') + \text{vector}('woman') \approx \text{vector}('queen')$

$\text{vector}('Paris') - \text{vector}('France') + \text{vector}('Italy') \approx \text{vector}('Rome')$

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 $f_{\text{synonym-of-}\langle \text{ngram } Z \rangle, \text{Plag.}}(x_1, x_2) =$
 $\text{get_similarity_with_embeddings()}$

Think-Pair-Share: Embedding Similarity

<https://vectors.nlpl.eu/explore/embeddings/en/>

These embeddings are created from Wikipedia. Consider how the words that are similar to them might have been calculated. That is, what articles do you think the words were found in?

Vector Embeddings: Key Ideas

Vector embeddings can be used for phrases, paragraphs, or even whole documents!

1. Acquire basic contextual statistics (often counts) for each word type v

2. Extract a real-valued vector e_v for each word v from those statistics

[0.00315225, 0.00315225, 0.00547597, 0.00741556, 0.00912817, 0.01068435, 0.01212381, 0.01347162, 0.01474487, 0.0159558]

3. Use the vectors to represent each word in later tasks

Shared Intuition Across Common Embedding Types

Model the meaning of a word by “embedding” in a vector space

The meaning of a word is a vector of numbers

Contrast: word meaning is represented in many computational linguistic applications by a vocabulary index (“word number 545”) or the string itself

Three Common Kinds of Embedding Models

- ~~1. Co-occurrence matrices~~
- ~~2. Matrix Factorization: Singular value decomposition/Latent Semantic Analysis, Topic Models~~
3. Neural-network-inspired models (skip-grams, CBOW)

Word2Vec

Mikolov et al. (2013; NeurIPS): “Distributed Representations of Words and Phrases and their Compositionality”

Revisits the context-word approach

Learn a model $p(c \mid w)$ to predict a context word from a target word

Word2Vec

Mikolov et al. (2013; NeurIPS): “Distributed Representations of Words and Phrases and their Compositionality”

Revisits the context-word approach

Learn a model $p(c \mid w)$ to predict a context word from a target word

Learn two types of vector representations

- $h_c \in \mathbb{R}^E$: vector embeddings for each context word
- $v_w \in \mathbb{R}^E$: vector embeddings for each target word

$$p(c \mid w) \propto \exp(h_c^T v_w)$$

Word2Vec

context (↓)-**word** (→) count matrix

	apricot	pineapple	digital	information
aardvark	0	0	0	0
computer	0	0	2	1
data	0	10	1	6
pinch	1	1	0	0
result	0	0	1	4
sugar	1	1	0	0

Context: those other words within a small “window” of a target word

$$\max_{h,v} \sum_{c,w \text{ pairs}} \text{count}(c, w) \log p(c | w)$$

$$p(c | w) \propto \exp(h_c^T v_w)$$

The wide road shimmered in the hot sun.

`tf.keras.preprocessing.sequence.skipgrams`



(wide, road)	...	(road, shimmered)	(hot, sun)	...	(the, hot)
(2, 3)	...	(3, 4)	(6, 7)	...	(1, 6)

`tf.random.log_uniform_candidate_sampler`
(negative_samples = 4)



(wide, road)	(wide, sun)	(wide, hot)	(wide, temperature)	(wide, code)
(2, 3)	(2, 7)	(2,6)	(2, 23)	(2, 2196)



concat and add label (pos:1/neg:0)



(wide, road)	(wide, sun)	(wide, hot)	(wide, temperature)	(wide, code)
(2, 3)	(2, 7)	(2,6)	(2, 23)	(2, 2196)
1	0	0	0	0

build context words and labels for all vocab words



Word	Context words					⇒	Labels				
2	3	7	6	23	2196	⇒	1	0	0	0	0
23	12	6	94	17	1085	⇒	1	0	0	0	0
84	784	11	68	41	453	⇒	1	0	0	0	0
							⋮				
V	45	598	1	117	43	⇒	1	0	0	0	0

- Positive example from training data
- Several negative examples (pairing two random words in the vocabulary)

<https://www.tensorflow.org/text/tutorials/word2vec>

Word2Vec has Inspired a Lot of Work

Off-the-shelf embeddings

- <https://code.google.com/archive/p/word2vec/>

Off-the-shelf implementations

- <https://radimrehurek.com/gensim/models/word2vec.html>

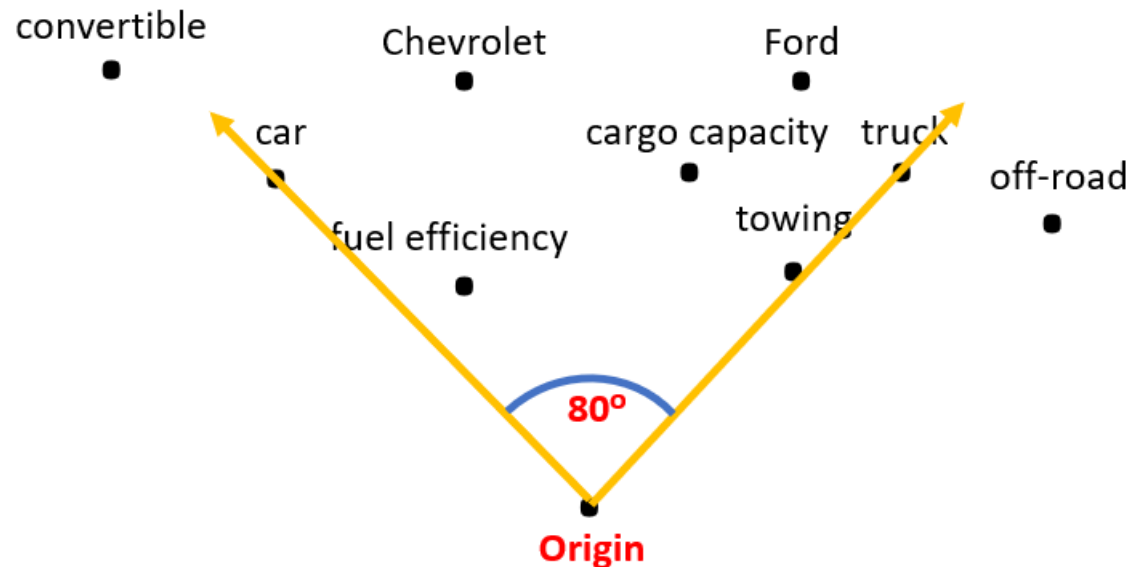
Follow-on work

- J. Pennington, R. Socher, and C. D. Manning, “**GLoVe: Global Vectors for Word Representation**,” in *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Doha, Qatar, 2014, pp. 1532–1543. doi: [10.3115/v1/D14-1162](https://doi.org/10.3115/v1/D14-1162).
 - <https://nlp.stanford.edu/projects/glove/>
- Many others
- 15000+ citations

Comparing/Evaluating Word Embeddings

Cosine: Measuring Similarity

Given 2 target words v and w how similar are their vectors?



<https://upload.wikimedia.org/wikipedia/commons/2/23/CosineSimilarity.png>

Cosine: Measuring Similarity

Given 2 target words v and w how similar are their vectors?

Dot product or inner product from linear algebra

$$\text{dot-product}(\vec{v}, \vec{w}) = \vec{v} \cdot \vec{w} = \sum_{i=1}^N v_i w_i = v_1 w_1 + v_2 w_2 + \dots + v_N w_N$$

- High when two vectors have large values in same dimensions, low for orthogonal vectors with zeros in complementary distribution

Correct for high magnitude vectors

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

Cosine Similarity

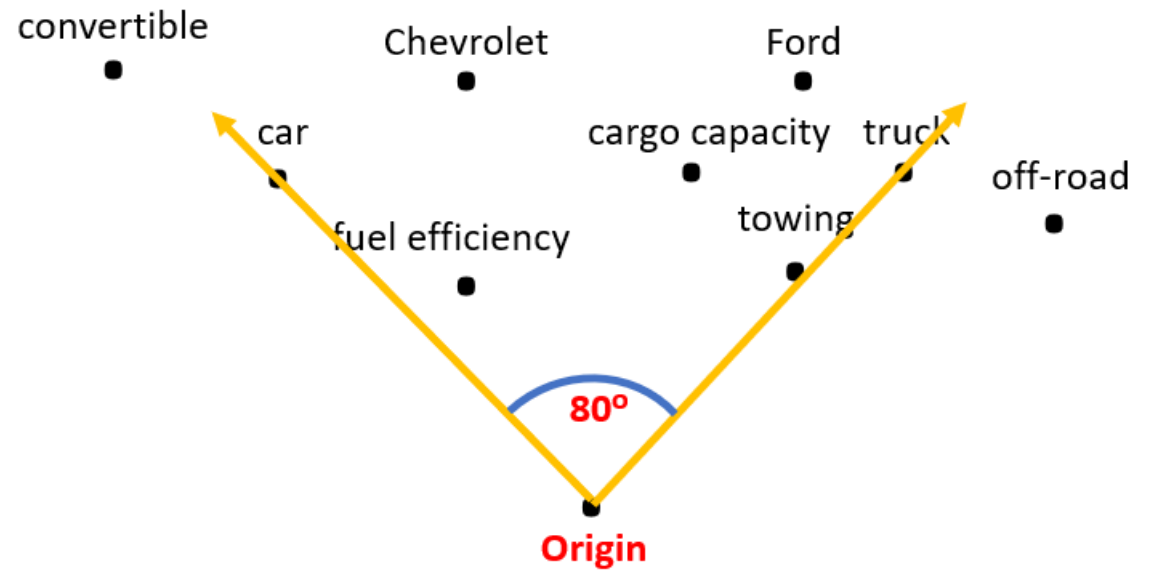
Divide the dot product by the length of the two vectors

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

This is the cosine of the angle between them

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \cos \theta$$



<https://upload.wikimedia.org/wikipedia/commons/2/23/CosineSimilarity.png>

Example: Word Similarity

$$\cos(x, y) = \frac{\sum_i x_i y_i}{\sqrt{\sum_i x_i^2} \sqrt{\sum_i y_i^2}}$$

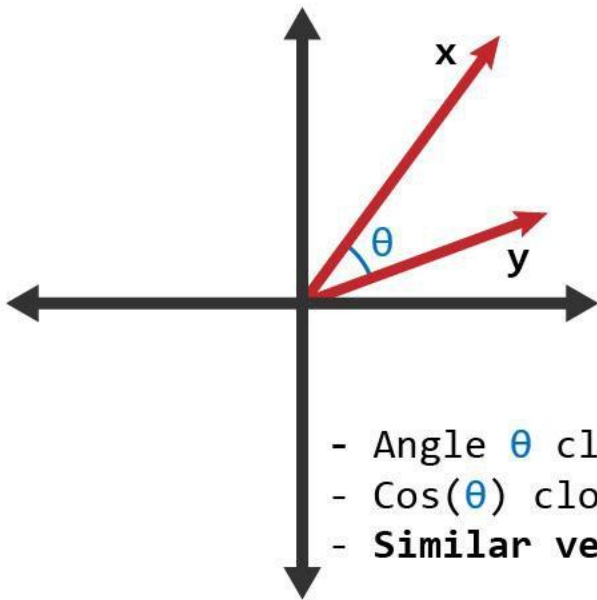
	Dim. 1	Dim. 2	Dim. 3
apricot	2	0	0
digital	0	1	2
information	1	6	1

$$\text{cosine}(\text{apricot}, \text{information}) = \frac{2 + 0 + 0}{\sqrt{4 + 0 + 0} \sqrt{1 + 36 + 1}} = 0.1622$$

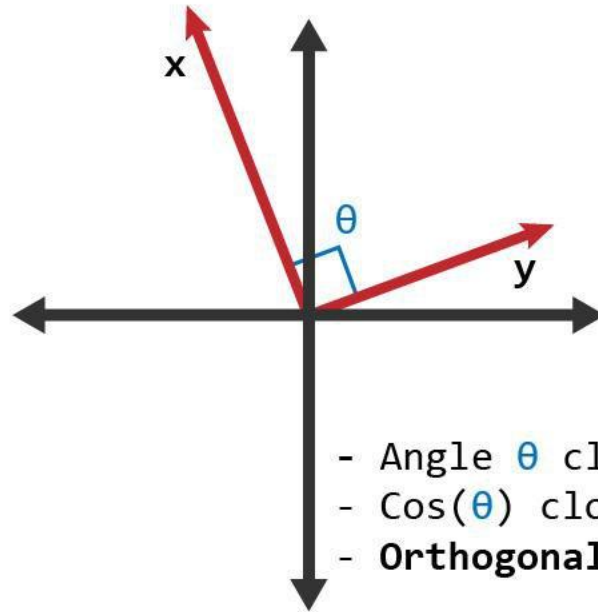
$$\text{cosine}(\text{digital}, \text{information}) = \frac{0 + 6 + 2}{\sqrt{0 + 1 + 4} \sqrt{1 + 36 + 1}} = 0.5804$$

$$\text{cosine}(\text{apricot}, \text{digital}) = \frac{0 + 0 + 0}{\sqrt{4 + 0 + 0} \sqrt{0 + 1 + 4}} = 0.0$$

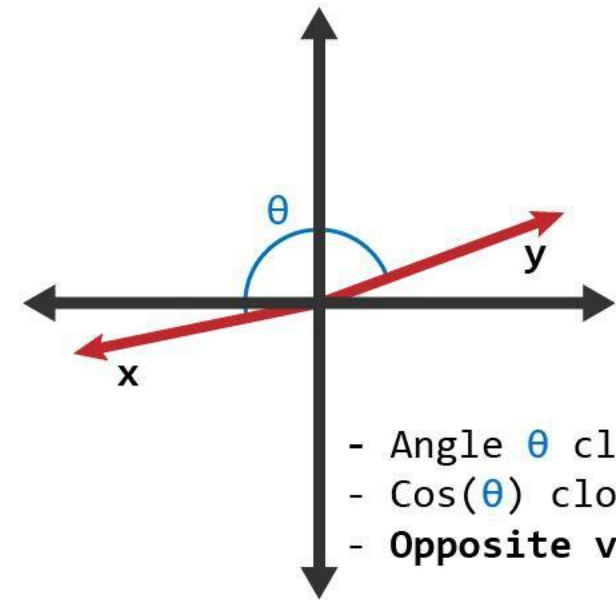
Cosine Similarity Range



- Angle θ close to 0
- $\cos(\theta)$ close to 1
- **Similar vectors**



- Angle θ close to 90
- $\cos(\theta)$ close to 0
- **Orthogonal vectors**



- Angle θ close to 180
- $\cos(\theta)$ close to -1
- **Opposite vectors**

<https://www.learnatasci.com/glossary/cosine-similarity/>

Other Neural Word Embedding Models

FastText

“Enriching Word Vectors with Subword Information” Bojanowski et al. (2017; TACL)

Main idea: learn **character n-gram embeddings** for the target word (not context) and modify the word2vec model to use these

Pre-trained models in 150+ languages

- <https://fasttext.cc>

FastText Details

Main idea: learn **character n-gram embeddings** and for the target word (not the context) modify the word2vec model to use these

Original word2vec:

$$p(c | w) \propto \exp(h_c^T v_w)$$

FastText:

$$p(c | w) \propto \exp\left(h_c^T \left(\sum_{\text{n-gram } g \text{ in } w} z_g\right)\right)$$

FastText Details

Main idea: learn **character n-gram embeddings** and for the target word (not the context) modify the word2vec model to use these

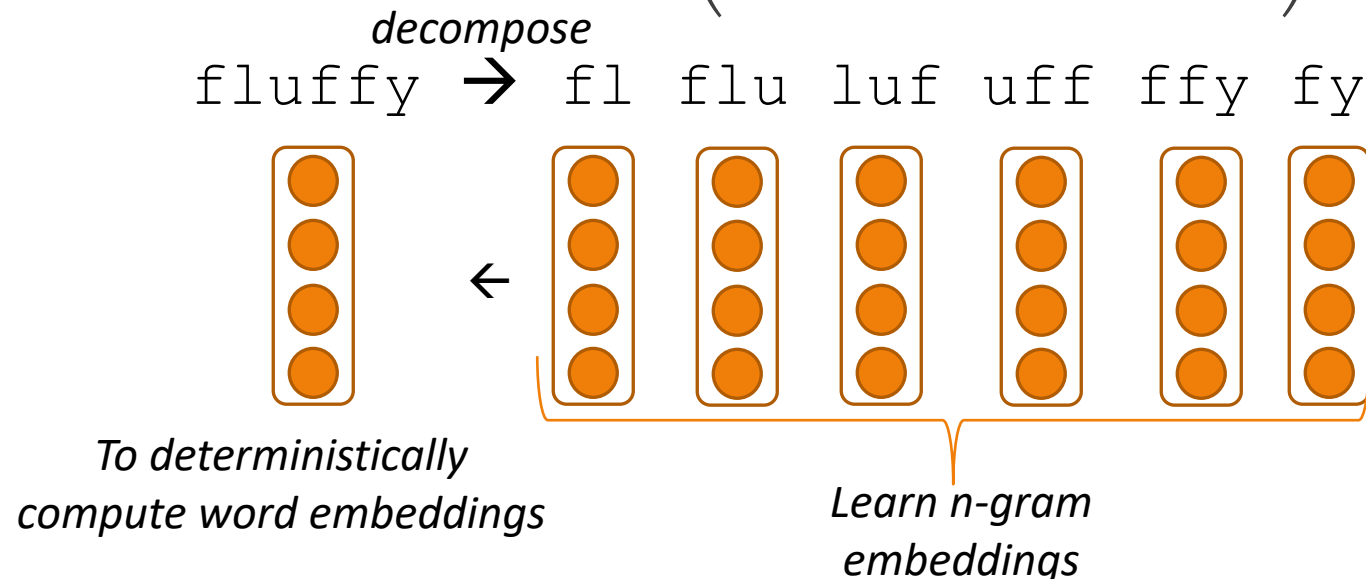
$$p(c | w) \propto \exp \left(h_c^T \left(\sum_{\text{n-gram } g \text{ in } w} z_g \right) \right)$$

decompose
fluffy $\rightarrow$ fl flu luf uff ffy fy

FastText Details

Main idea: learn **character n-gram embeddings** and for the target word (not the context) modify the word2vec model to use these

$$p(c | w) \propto \exp \left(h_c^T \left(\sum_{\text{n-gram } g \text{ in } w} z_g \right) \right)$$



Contextual Word Embeddings

Word2vec-based models are not context-dependent

Single word type → single word embedding

If a single word type can have different meanings...

bank, bass, plant,...

... why should we only have one embedding?

Entire task devoted to classifying these meanings:

Word Sense Disambiguation

Contextual Word Embeddings

Growing interest in this

Off-the-shelf is a bit more difficult

- Download and run a model
- Can't just download a file of embeddings

Two to know about (with code):

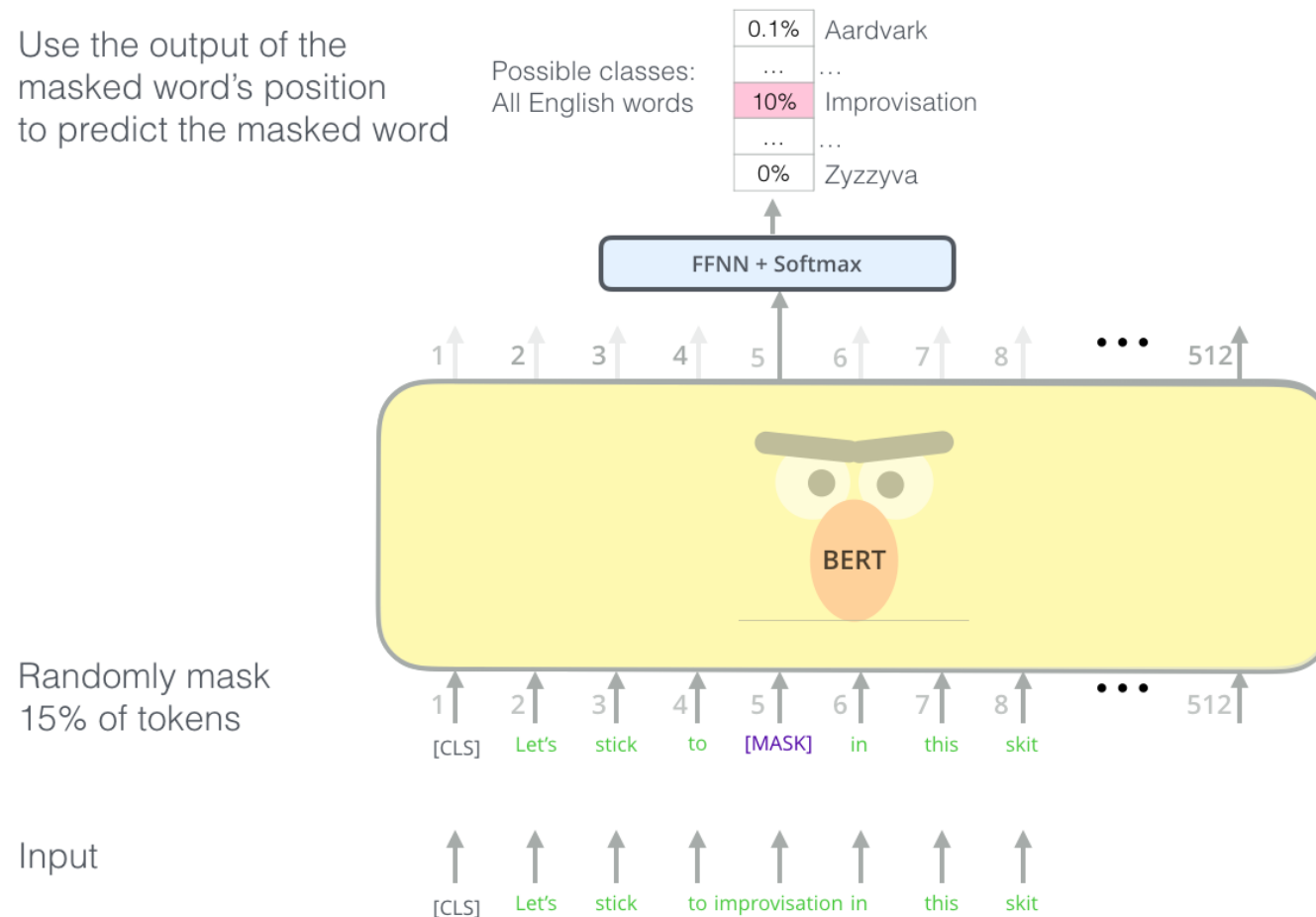
- ELMo: “Deep contextualized word representations” Peters et al. (2018; NAACL)
- <https://allennlp.org/elmo>
- BERT: “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding” Devlin et al. (2019; NAACL)
- <https://github.com/google-research/bert>



Back to Transformers...

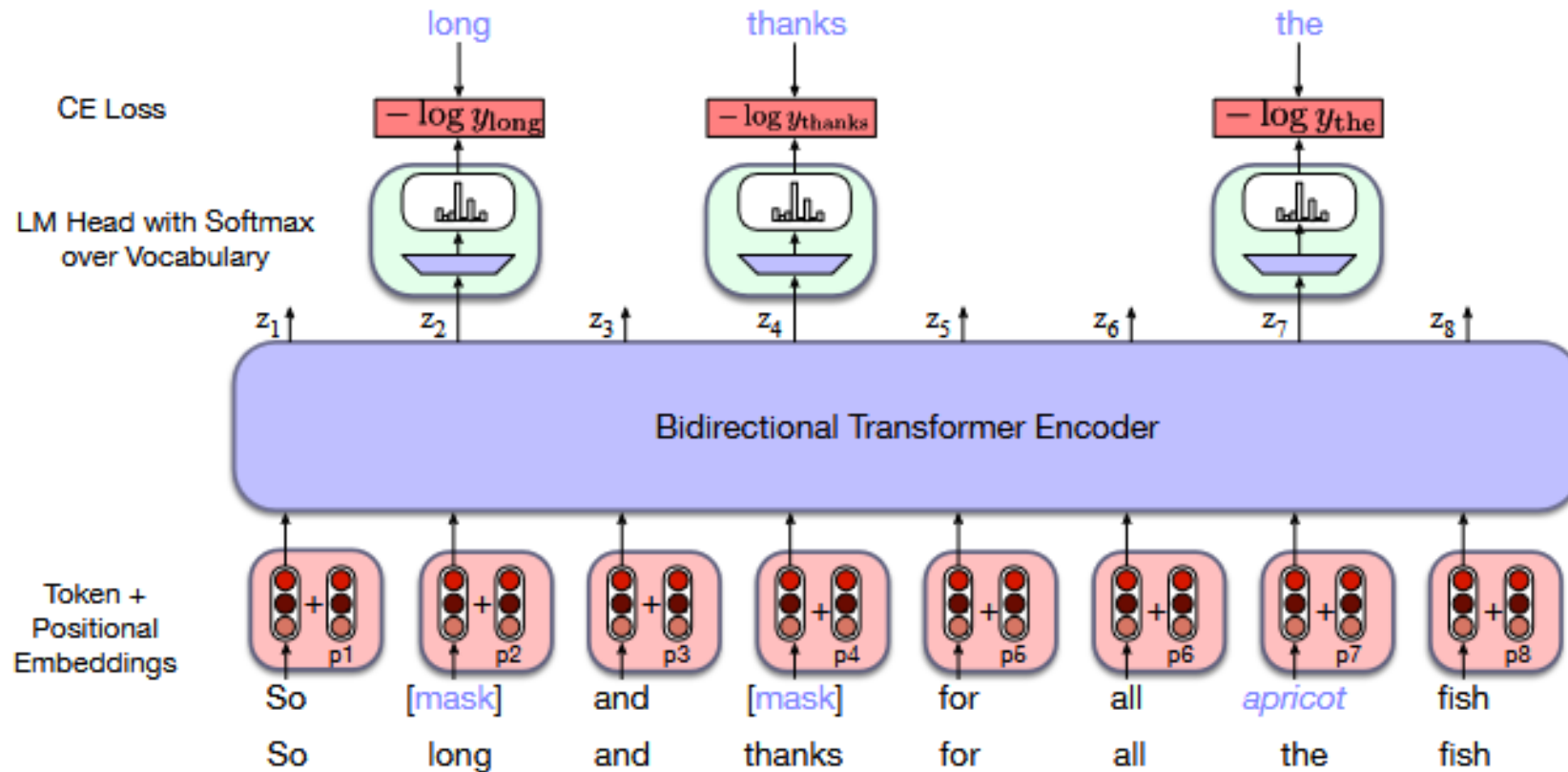
BERT (Devlin et al. 2019)

Use the output of the masked word's position to predict the masked word



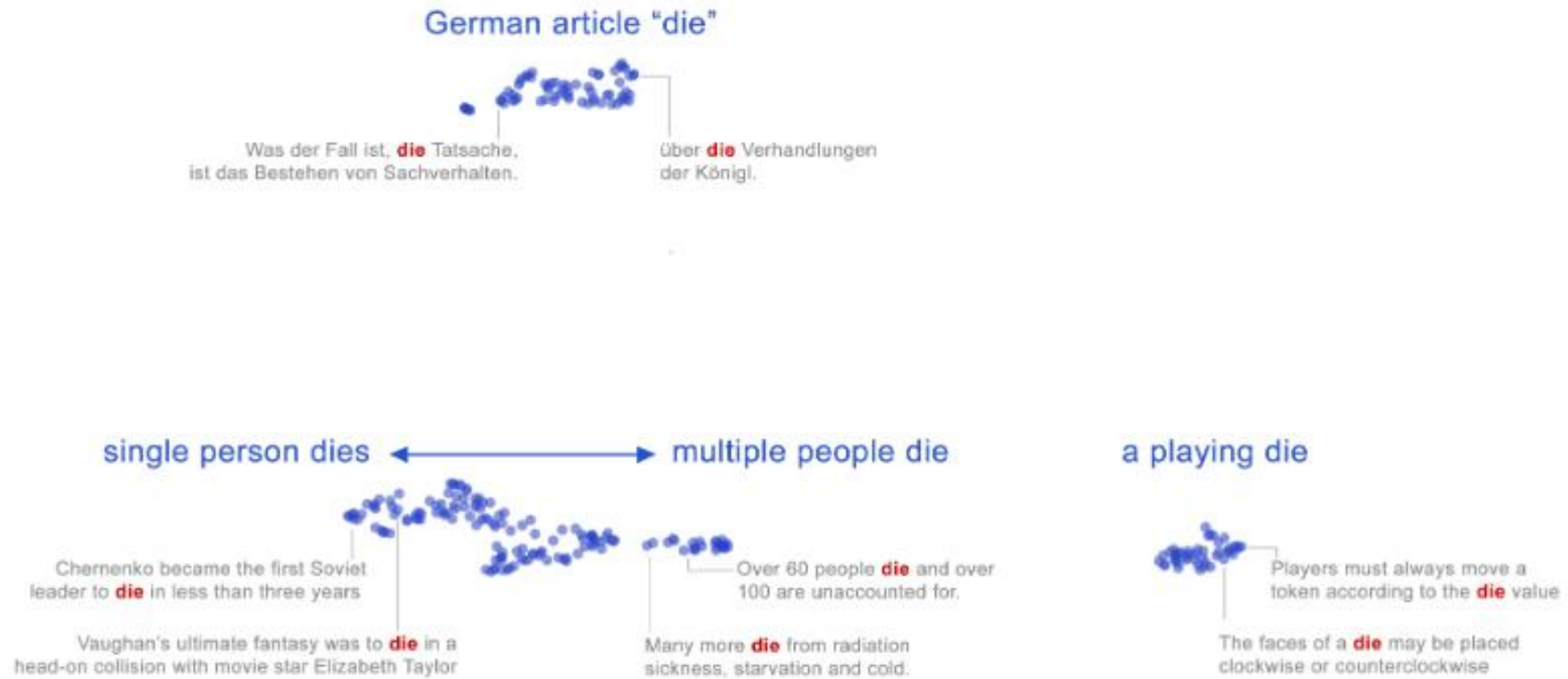
<http://jalammr.github.io/illustrated-bert/>

Masked Language Models



From Jurafsky & Martin's *Speech and Language Processing*, 3rd Edition, Chapter 11

Contextual Embeddings



From Jurafsky & Martin's *Speech and Language Processing*, 3rd Edition, Chapter 11

Uses of Encoder-Only Models

Classification tasks

Sentence embeddings

Context-dependent word embeddings

Any type of fill-in-the-blank tasks

BERT Question

Consider the highlighted words. Which two words would contextual word embeddings from BERT say are closest?

- A. I am so excited to use my new bat at the baseball game tomorrow.
- ✓ B. The favorite food of this species of bat is mosquitoes.
- ✓ C. The cardinal isn't just a lawn decoration; the species makes themselves useful by eating mosquitoes.

Remember: word2vec is a dense vector embedding

Word2Vec Question

Consider the highlighted words. Which two words would word2vec say are closest?

- ✓ A. I am so excited to use my new bat at the baseball game tomorrow.
- ✓ B. The favorite food of this species of bat is mosquitoes.
- C. The cardinal isn't just a lawn decoration; the species makes themselves useful by eating mosquitoes.

BERT Family of Models

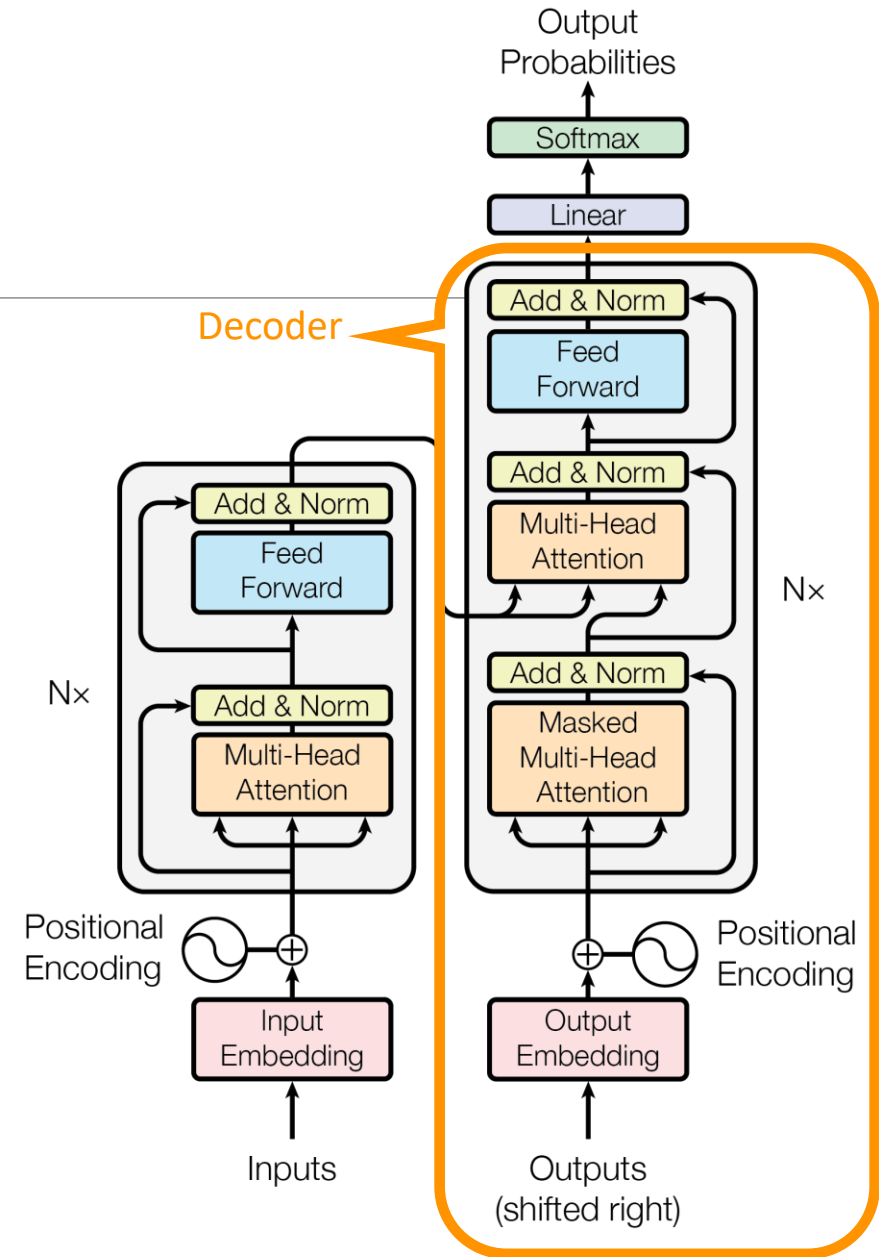
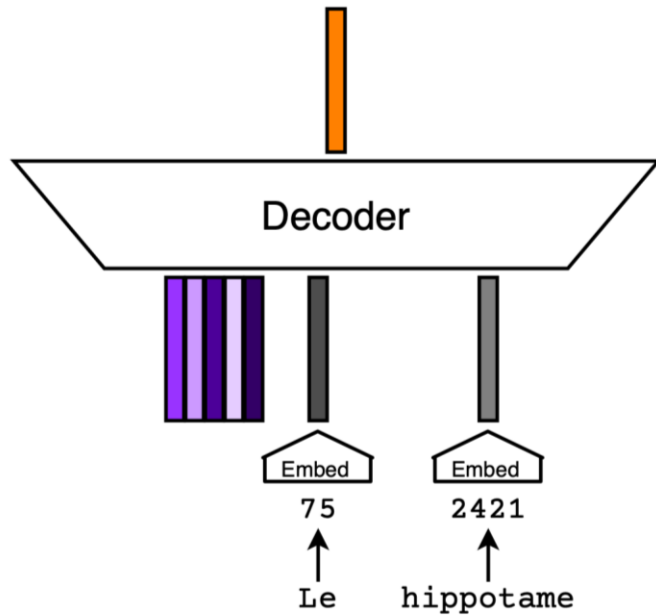
- Encoder-only
 - Input: Corrupted version of text sequence
 - Goal: Produce an uncorrupted version of text sequence
- How to use:
 - Finetune for a classification task
 - Extract word/sentence embeddings

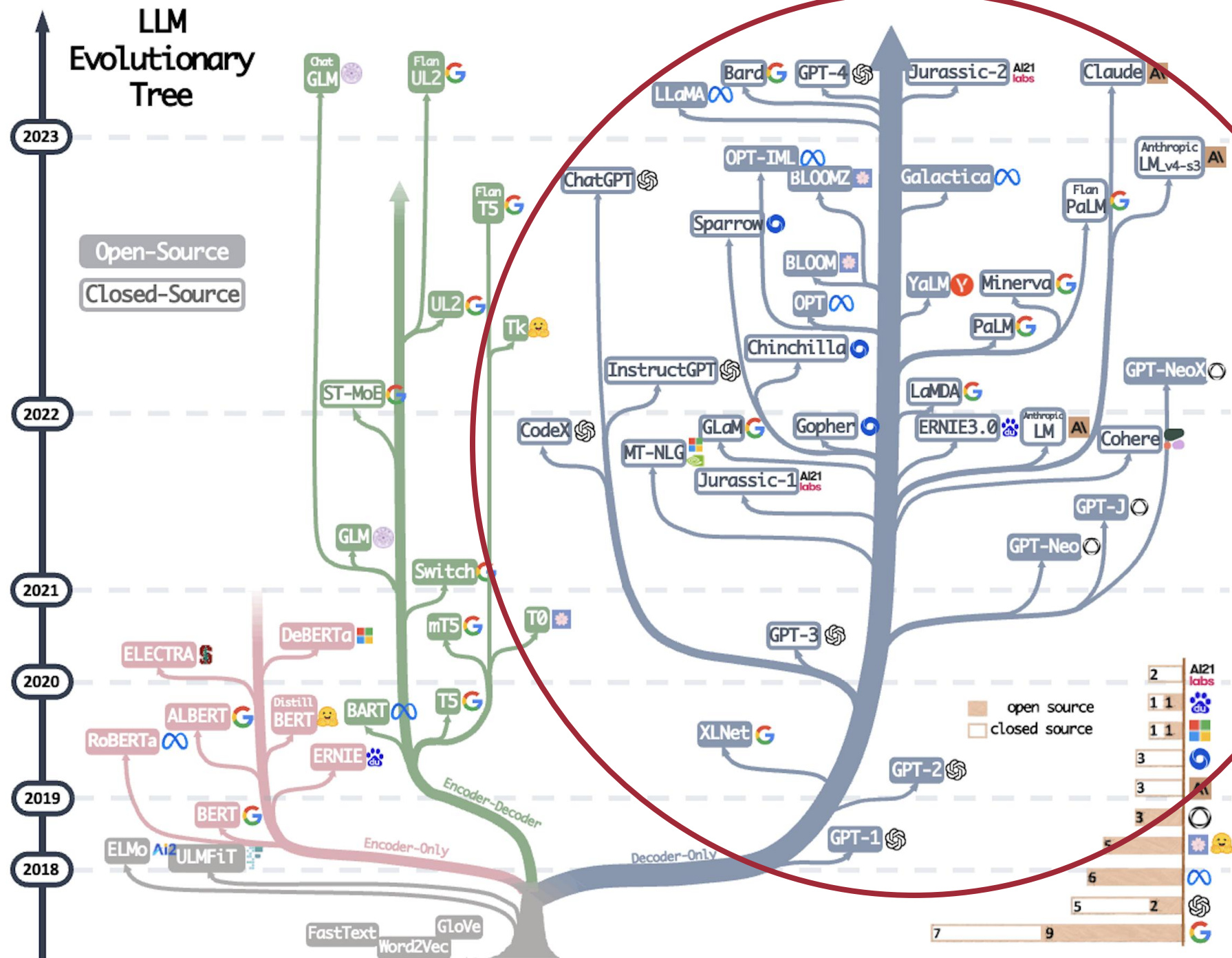
Some important BERT family members

(in my opinion)

- RoBERTa (better version of the original BERT) – Liu et al. 2019 (Facebook)
- Sentence-BERT (BERT fine-tuned to give good sentence embeddings) – Reimers & Gurevych 2019 (Technische Universität Darmstadt)
- DistilBERT (lite BERT) – Sanh et al. 2019
- ALBERT (lite BERT) – Lan et al. 2020
- HuBERT (BERT for speech embeddings) – Hsu et al. 2021

Decoder-Only Models





GPT Family

- Decoder-only
 - Input: Text sequence
 - Goal: Predict the next word given the previous ones
- How to use:
 - Ask GPT* to continue from a prompt.
 - Finetune smaller GPTs for more customized generation tasks.
 - ChatGPT cannot be finetuned since it is already finetuned
 - Use OpenAI's API to get them to fine-tune GPT* for you.
- Around GPT-2 was when pre-trained models became popular
- Around GPT-3 was when *just* prompting became a thing

Other Decoder-Only Models

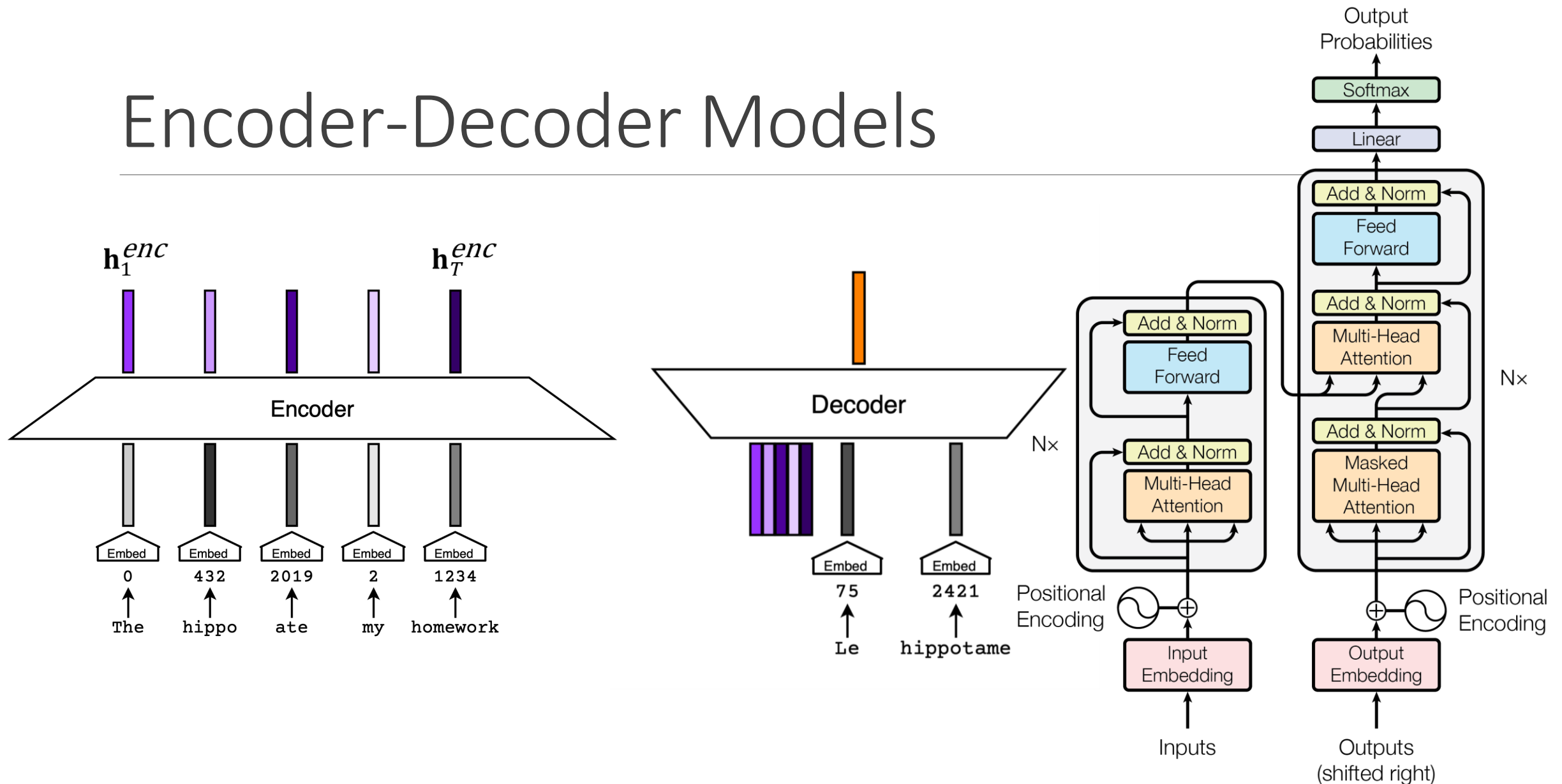
LLaMA 3/4 (Meta)

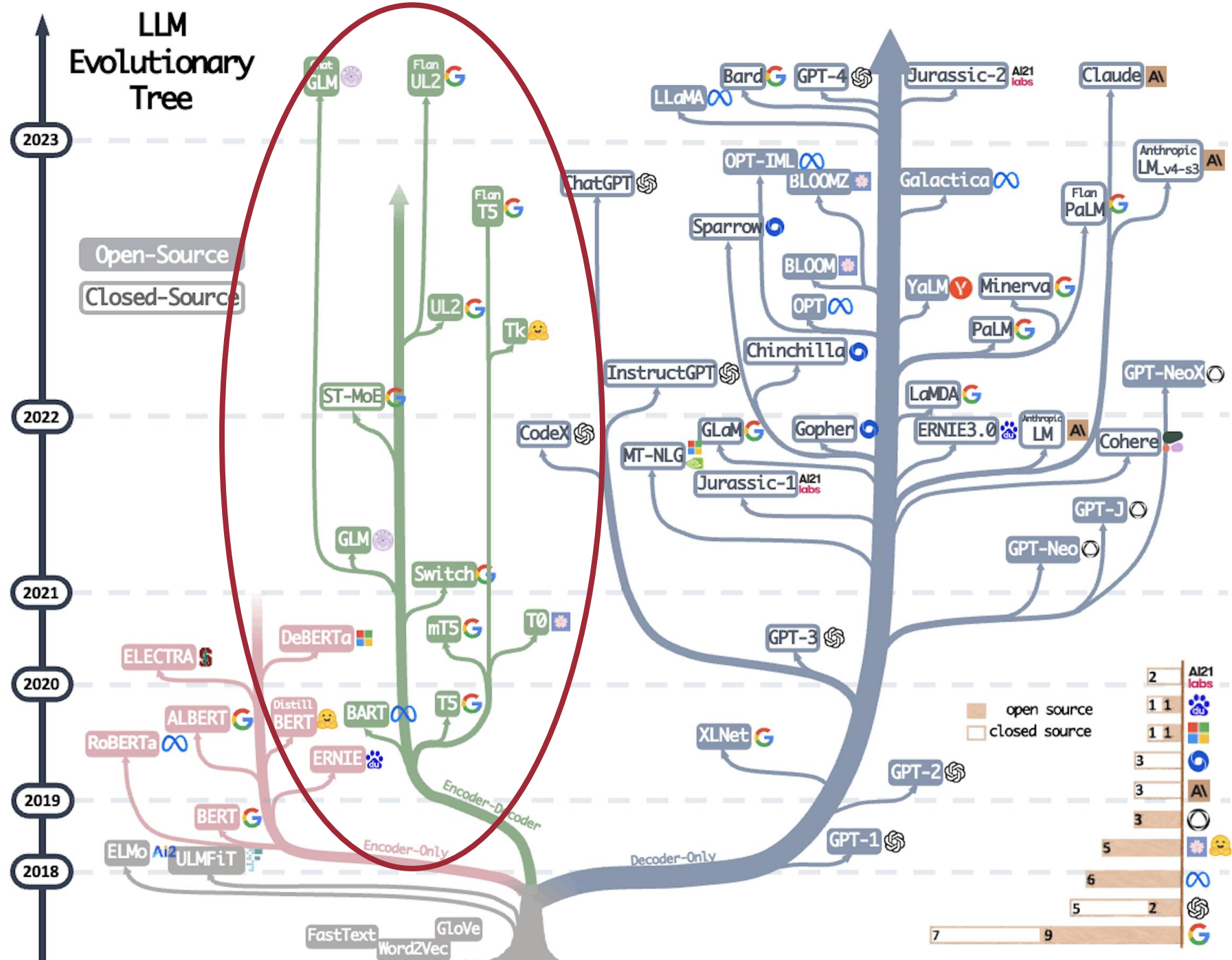
Claude 3 (Anthropic)

Gemma (Google)

OLMo 2 (AI2)

Encoder-Decoder Models



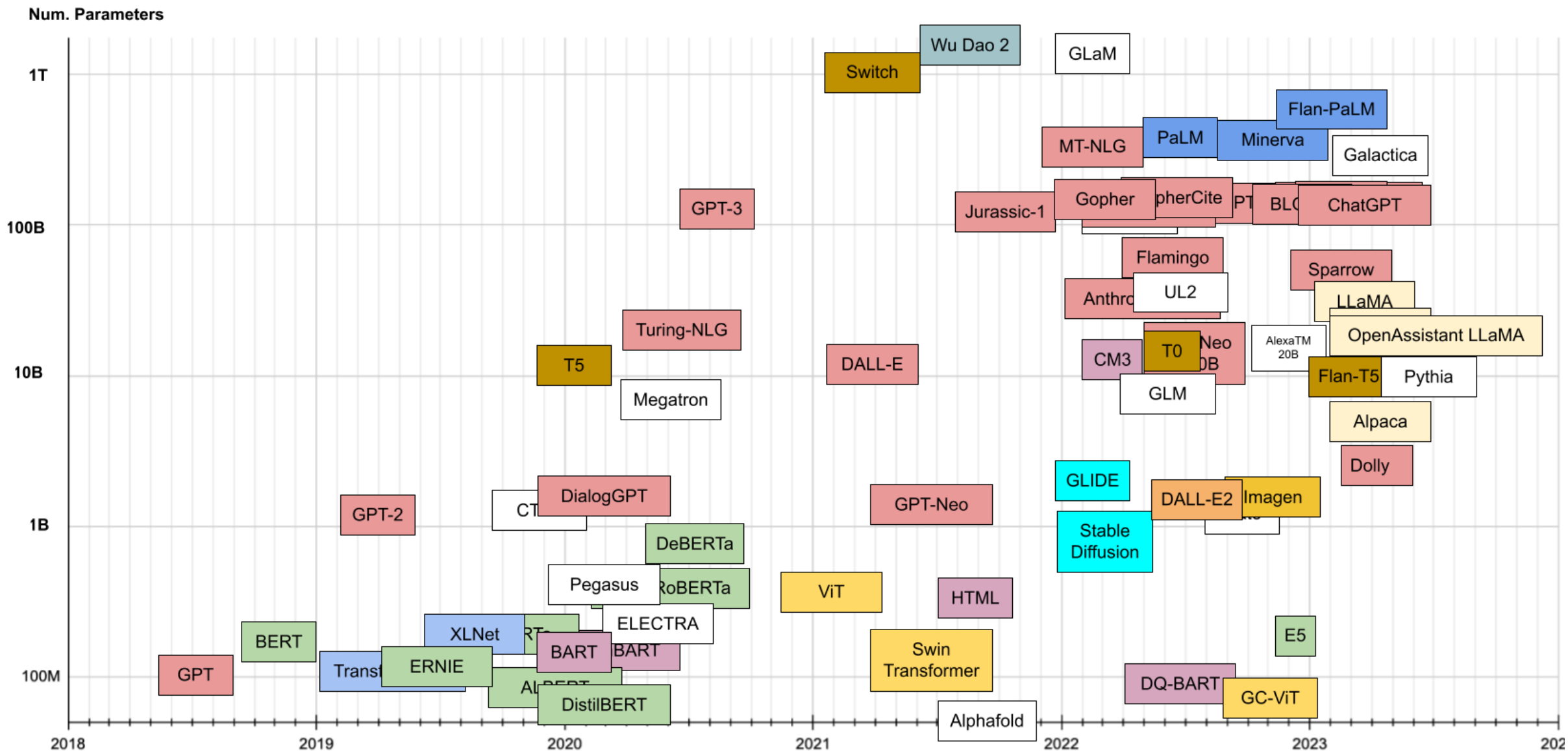


Enc-Dec Family of Models

- Encoder-decoder
 - Input: Text sequence with random word spans deleted
 - Goal: Generate the deleted word spans
- How to use:
 - Finetune smaller ones for either generation or classification tasks.
 - Prompt tuning (train a sequence of embedding which get prefixed to the input)

Some Enc-Dec family members

- T5 – Raffel et al. 2020 (Google)
- BART (combo of GPT and BERT) – (Facebook)
- DALL-E 2 (for caption prediction)



<https://amatriain.net/blog/transformer-models-an-introduction-and-catalog-2d1e9039f376/>